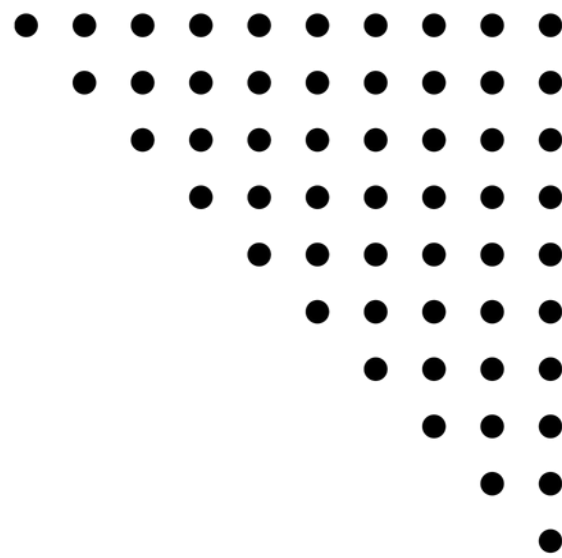


AI-Assisted Predictive Maintenance and Smart Control of Power Converters Using F28379D and Raspberry Pi



Final Year Project:

**Coochbehar Government Engineering College
Cooch Behar, West Bengal
2026**

Electronics and Communication Engineering

Presented by :

- 1. Ankit KumarOjha (34900322043)**
- 2. Arpan Mukherjee (34900322019)**
- 3. Pallab Acharjee (34900322009)**
- 4. Sourav Ghosh (34900322014)**
- 5. Subham Mandal (34900322022)**
- 6. Sneha Saha (34900323064)**

Project Guide : Dr. Sourav Chakraborty



Acknowledgement

The authors express their sincere gratitude to the Department of Electronics and Communication Engineering, Cooch Behar Government Engineering College, for providing the opportunity and necessary facilities to carry out this project work. We would like to thank our Project Supervisor for their valuable guidance, encouragement, and technical support throughout the duration of this project. We are also thankful to the Technical Facilitator for their continuous assistance and constructive suggestions. We extend our sincere thanks to all faculty members of the department for their cooperation and support. Finally, we express our heartfelt gratitude to our family members and friends for their constant motivation and encouragement during the completion of this project



AGENDA

- **Introduction**
- **Motivation**
- **Objectives**
- **DC–DC Buck Converter**
- **Buck Converter Working**
- **Hardware Components**
- **System Block Diagram**
- **Working Principle**
- **Circuit Diagram**
- **Software Flow**
- **Hardware Implementation**
- **Experimental Results**
- **Applications**
- **Advantages**
- **Future Scope**
- **Conclusion**



Introduction

- Power converters efficiently convert electrical energy from one voltage level to another.
- A DC–DC buck converter reduces higher DC voltage to a lower regulated voltage.
- Digital control using the C2000 improves accuracy, flexibility, and monitoring compared with conventional analog controllers.



Motivation

- Need for efficient and programmable DC power supplies.
- Digital control provides precise voltage regulation.
- Real-time monitoring improves converter performance.
- Practical implementation enhances understanding of embedded systems and power electronics.
- Suitable for laboratory experiments and educational applications.



Objectives

- Design a digitally controlled buck converter.
- Generate PWM using C2000 DSP.
- Regulate output voltage from 6–28 V input.
- Measure voltage and current continuously.
- Display operating parameters on OLED.
- Adjust output using a rotary encoder.

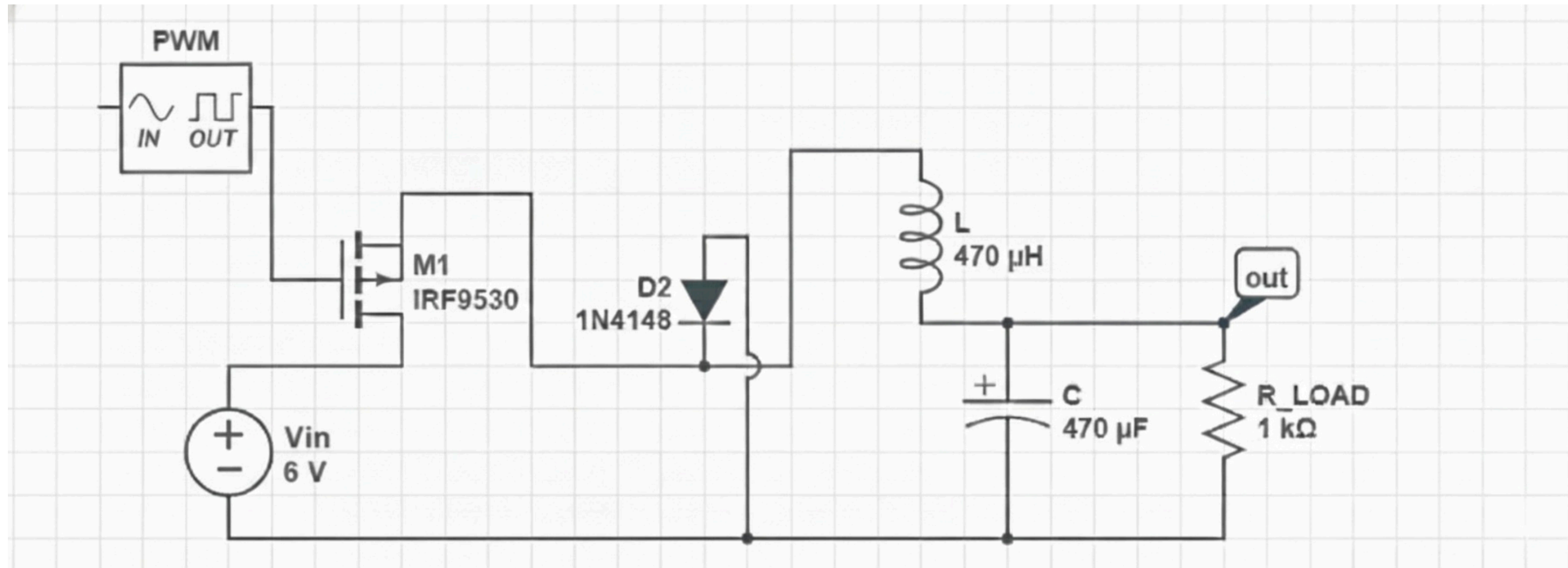


DC-DC Buck Converter

- A buck converter is a step-down switching converter.
- Converts higher DC voltage into lower regulated DC voltage.
- Uses PWM switching for voltage control.
- Provides high efficiency and low power loss.
- Widely used in battery-powered and embedded systems.



Buck Converter Circuit Diagram



Buck Converter Working

- MOSFET switches at high frequency.
- Inductor stores energy during ON time.
- Diode conducts during OFF time.
- Capacitor filters voltage ripple.
- Output voltage depends on PWM duty cycle generated by the DSP.

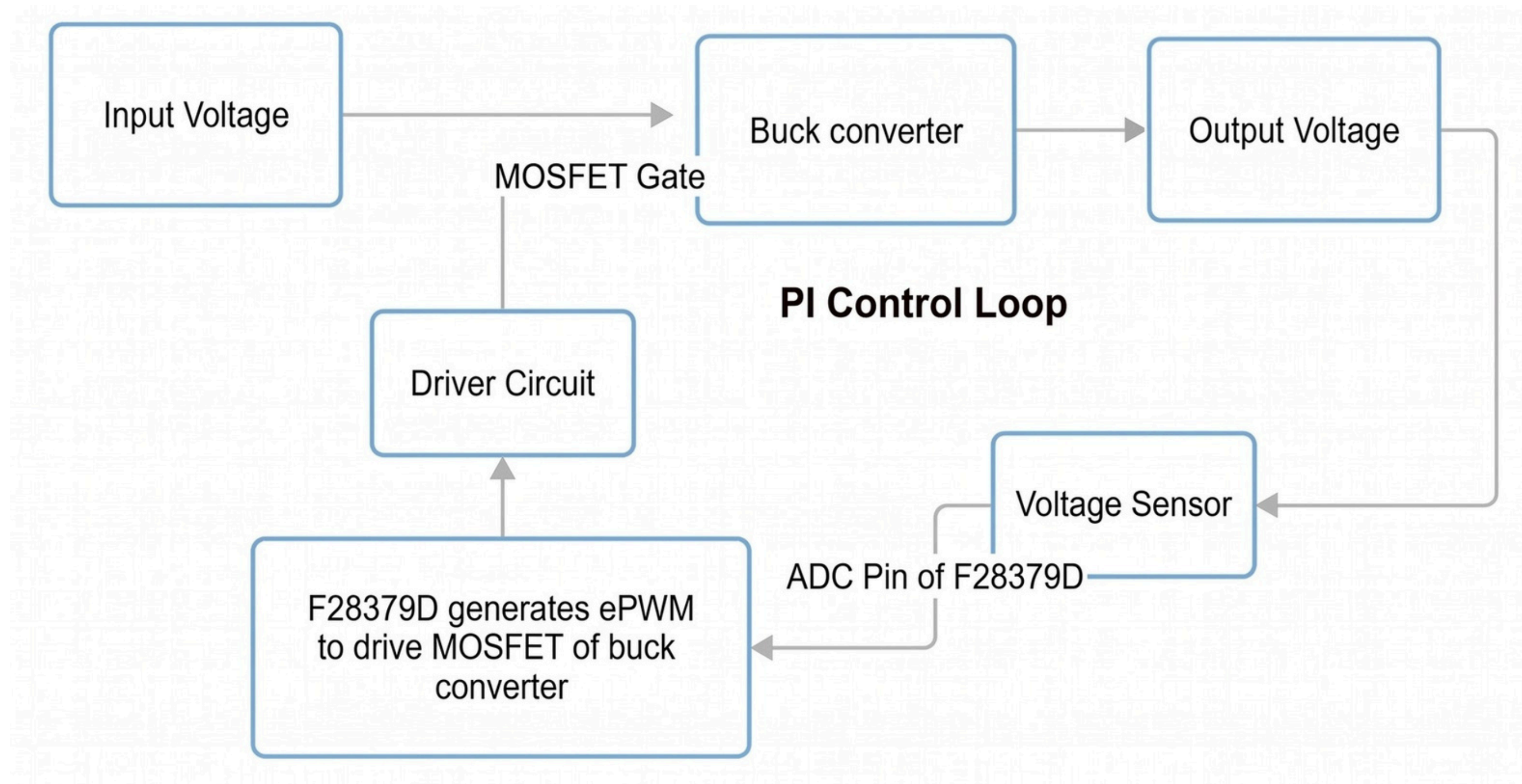


Hardware Components

- C2000
- ESP32
- MP2388 Power Module
- TC4420 Gate Driver
- IRLZ44N MOSFET
- INA240 Current Sensor
- Voltage Divider
- Rotary Encoder
- OLED Display
- 220 μ H Inductor
- Freewheeling Diode
- Output Capacitor



System Block Diagram

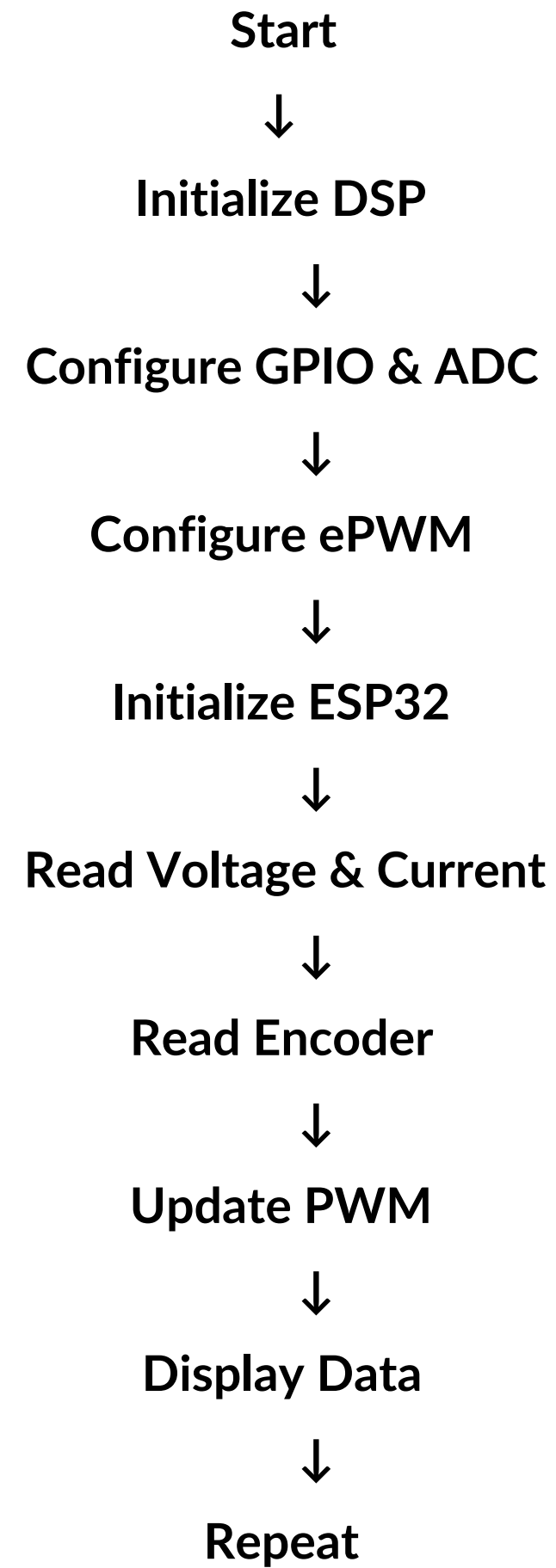


Working Principle

- InputSupply:6–28 VDC
- MP2388 provides regulated 5 V to control circuits.
- DSP reads voltage and current through ADC.
- Rotary encoder sets the desired output voltage.
- DSP generates PWM using ePWM1A.
- TC4420 drives IRLZ44N MOSFET.
- Inductor and capacitor smooth the output.
- Output: Variable regulated DC voltage



Software Flow



Applications

- Laboratory DC power supplies
- Battery charging systems
- Embedded electronic systems
- Renewable energy applications
- Industrial automation
- Educational power electronics experiments
- Low-voltage control systems



Advantages

- High efficiency
- Digital PWM control
- Accurate voltage regulation
- Fast MOSFET switching
- Compact hardware
- User-adjustable output
- Real-time monitoring
- Easy software modification



Future Scope

- Implement PID control.
- Add wireless monitoring using ESP32 Wi-Fi.
- Increase power rating.
- Improve protection circuits.
- Develop a programmable power supply.
- Extend to Boost and Buck–Boost converters.
- Add data logging capability.



Conclusion

- Successfully implemented a DSP-based DC–DC buck converter.
- Achieved stable voltage regulation using digital PWM.
- Integrated sensing, display, and user control.
- Experimental results validated hardware and software performance.
- The project provides a practical platform for digital power electronics.

